THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 80°	i + r = 40 + 40.
2	**(b)** 20 m	Ratio 2:1.
3	**(b)** half	Inversely.
4	**(b)** 65°	90 - 25.
5	**(d)** noon	Sun overhead.
6	**(b)** away from the hole	Longer distance.
7	**(a)** 0°	Retraces its path.
8	**(a)** sunrise or sunset	Sun low.
9	**(a)** blurred and brighter	Wide beam.
10	**(c)** 2 m	1 m from the mirror, so 2 m.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reflection 35° (1); Between rays 70° (1).	1+1
12	It reflects more light (1); Image is bright (1).	1+1
13	Two reflections (1); Image remains upright (1).	1+1
14	New distance 1 m from mirror (1); Image 1 m behind: 2 m apart (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Incidence = 90 - 70 = 20° (1); Reflection 20° (1); Between rays 40° (1).	1+1+1
16	Shows the road ahead (1); Helps avoid accidents (1); Wide view (1).	1+1+1
17	Regular reflection in mirror (1); Diffused in wall (1); Rough surface scatters (1).	1+1+1
	OR: Height 1.4 m (1); Image 3 m behind the mirror (1); Distance from her 6 m (1).	1+1+1
18	Smooth surface gives regular reflection (1); Ripples scatter light (1); Image is distorted (1).	1+1+1
	OR: Look above water (1); While submerged (1); Two mirrors (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Normal drawn (1); Angle of incidence 30° (1); Angle of reflection 30° (1); Angle between incident and reflected 60° (1); Reflected ray on opposite side of the normal (1).	1+1+1+1+1
	OR: Solar: Moon between Sun and Earth (1); Lunar: Earth between (1); Umbra and penumbra (1); Straight travel of light (1); Occurs rarely (1).	1+1+1+1+1
20	Closer object gives larger image (1); Farther object gives smaller (1); Larger screen distance gives larger image (1); Brightness decreases (1); Inverted always (1).	1+1+1+1+1
	OR: New distance from the mirror = 3 m (1); The image is 3 m behind the mirror (1); Distance between her and image = 6 m (1); Image distance equals object distance (1); The image is virtual (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** east	Sun in the west.
2	**(c)** the hole is covered	No light enters.
3	**(c)** 1.2 m tall	Same size.
4	**(a)** decreases	Fewer rays blocked.
5	**(b)** real image	Light actually reaches.
6	**(c)** 3 m	Half.
7	**(c)** decreases	Rays diverge less.
8	**(c)** a round spot	Pinhole effect.
9	**(a)** 60°	Twice.
10	**(d)** sharp shadow	Point source.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compare shadows of tree and stick (1); Ratio gives the height (1).	1+1
12	Prevent dazzling drivers (1); Controlled beams (1).	1+1
13	It reflects light from a source (1); The light enters our eyes (1).	1+1
14	Image is virtual (1); Light does not actually meet there (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Box with a hole and screen (1); Candle in front (1); Inverted image on the screen (1).	1+1+1
16	Object 2 m in front (1); Image 2 m behind (1); Equal distances (1).	1+1+1
17	To see the back of the teeth (1); The mirror reflects light into the mouth (1); The image shows hidden parts (1).	1+1+1
	OR: Sun is a point-like source (1); Clouds scatter light (1); Light comes from all directions, soft shadows (1).	1+1+1
18	Multiple images (1); See the back of the head (1); Repeated reflection (1).	1+1+1
	OR: Mirrors inclined (1); Multiple images (1); Symmetry (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tube with openings (1); Mirror at top at 45° (1); Mirror at bottom at 45° (1); Ray reflects twice (1); Eye sees erect image (1).	1+1+1+1+1
	OR: Non-luminous (1); Reflect sunlight (1); Enter eyes (1); Rough reflection (1); Visible from all directions (1).	1+1+1+1+1
20	Mirror on a white paper (1); Draw the normal (1); Place pins along incident and reflected (1); Measure the angles (1); Equal (1).	1+1+1+1+1
	OR: Three mirror strips (1); Tube and cap (1); Coloured beads (1); Reflections (1); Rotation shifts beads (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** crescent images of the Sun	Shape matches.
2	**(d)** MOM	M and O are symmetric letters.
3	**(a)** shadow with umbra and penumbra	Extended source.
4	**(c)** the colour of the object	Colour is irrelevant.
5	**(d)** 9:00	Lateral inversion.
6	**(c)** makes a faint shadow	Translucent.
7	**(a)** a sunny field	Light present.
8	**(a)** A	Symmetric letter.
9	**(a)** a perfectly transparent medium such as clean air	Light passes.
10	**(c)** many images of an object	360/60 - 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Subtract from 12:00 (1); 7:40 (1).	1+1
12	Sun's position in the sky changes (1); Shadow longer at morning and evening (1).	1+1
13	Seeing objects from all directions (1); Books on a page (1).	1+1
14	Make the room look bigger (1); Customers see themselves (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Closer to bulb: larger (1); Closer to screen: smaller (1); Shape stays similar (1).	1+1+1
16	Real image formed on screen (1); Pinhole image is real (1); Inverted (1).	1+1+1
17	A smooth water surface reflects regularly (1); Light bounces back in an orderly way (1); So an image is seen (1).	1+1+1
	OR: Light enters top mirror (1); Reflects down (1); Second mirror reflects into the eye (1).	1+1+1
18	White paper reflects (1); Ink absorbs (1); Contrast (1).	1+1+1
	OR: Very faint (1); Most light passes (1); Edges may cast faint (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dressing mirror (1); Rear-view mirror (1); Periscope (1); Still water (1); Torch reflectors (1).	1+1+1+1+1
	OR: Shadow lengthens (1); Then another lamp (1); Shadows rotate (1); Position of lamp matters (1); Light travels straight (1).	1+1+1+1+1
20	Large lamp has many points (1); Umbra gets no light (1); Penumbra partial (1); Edges blurred (1); Small lamp gives sharp edge (1).	1+1+1+1+1
	OR: Reflect sunlight (1); Concentrate on a spot (1); Pot heats (1); Shiny surface better (1); Increase efficiency (1).	1+1+1+1+1

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